Manwe Castro

manwe123@asu.edu | (602) 350-3164 | linkedin.com/in/manwecastro | https://manwe-123.github.io/

Education

Bachelor of Science in Mechanical Engineering Tempe, AZ
Arizona State University | GPA: 3.50 *May 2028*

Professional Experience

Institute for Digital Inclusion Acceleration Chandler, AZ
Digital Navigator *January 2025 – Present*

- Instruct over 300 weekly visitors in hardware and software fundamentals, facilitating thousands of hands-on technology interactions across diverse age groups.
- Develop and deliver structured curriculum for nine distinct software platforms, including Swift, TinkerCAD, and SpheroEDU, standardizing the instructional process for the center.
- Operate and maintain Flashforge 3D printers using Orca Slicer, troubleshooting hardware failures and optimizing print parameters to ensure consistent output quality.
- Mentor community members on digital literacy and device safety, translating complex technical concepts for users with varying levels of technical proficiency.

Project Experience

VR Rapid Prototyping Application Tempe, AZ
Prototyping Lead | Modeling Team *March 2025 – December 2025*

- Developed a virtual reality prototyping environment for Meta Quest headsets using the Unity engine, integrating over 100 interactive 3D assets.
- Directed a team of modelers in Unity, establishing technical workflows and resolving engine blockers to deliver the functional prototype by the end of the semester.
- Partnered with faculty to design the underlying curriculum framework, establishing the technical foundation for future integration into introductory engineering courses.

Documentation Lead *January 2025 – March 2025*

- Authored technical reports, Gantt charts, and meeting minutes to track design milestones and maintain project version control.
- Standardized reporting templates to improve progress tracking and delivered a final technical package detailing project scope, design rationale, and future development recommendations.

Mental Health Robotics System for Isolated Environments Tempe, AZ
Design and Prototyping Lead *August 2024 – December 2024*

- Programmed an Arduino Uno microcontroller to process inputs from ultrasonic distance and photoresistive light sensors, enabling semi-autonomous environmental navigation.
- Fabricated the physical robot chassis using laser-cut wooden components, ensuring precise mechanical fit and structural rigidity for the electronic modules.
- Executed successful demonstration trials of the semi-autonomous prototype, validating sensor accuracy and mechanical stability under operational conditions.

Computer-Aided Design Modeling Projects *August 2025 – December 2025*

- Modeled complex mechanical assemblies in SolidWorks and Fusion 360, applying design intent principles to ensure manufacturability and proper part fit.
- Generated fully dimensioned 2D engineering drawings with standard drafting conventions, creating manufacturing-ready documentation for machined and 3D-printed components.

Skills

Technical: SolidWorks, Autodesk Fusion 360, MATLAB, TinkerCAD, Arduino Circuitry & Coding, Unity Game Engine, 3D Printing, Microsoft Suite, and Google Workspace